Lesson 5
GEOMETRIC TRANSFORMATIONS

2D Geometric Transformations

- What are geometric transformations?
  - Operations that are applied to the geometric description of an object to change its position, orientation, or size.
- Why the transformation is needed?
  - To manipulate the initially created object and to display the modified object without having to redraw it.
- Two ways:
  1. Object Transformation
     - Alter the coordinates descriptions of an object
     - Translation, rotation, scaling etc.
     - Coordinate system unchanged
  2. Coordinate transformation
     - Produce a different coordinate system

2D Geometric Transformations

- **Basic Transformations**
  - Translation
  - Rotation
  - Scaling
- **Other transformations**
  - Reflection
  - Shear

Translation

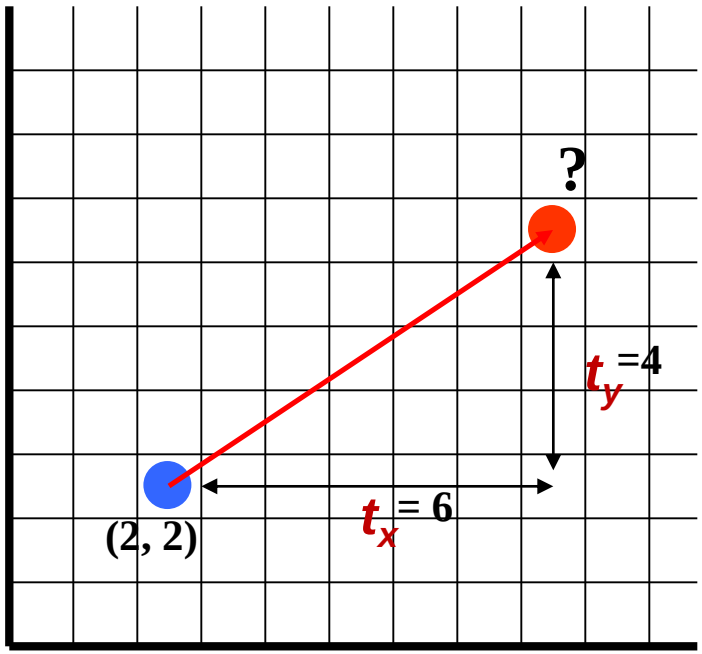
- A translation moves all **points** in an object along the same straight-line path to new **positions**.
- The path is represented by a vector, called the **translation** or **shift vector**.
- We can write the components:

$$p'_x = p_x + t_x$$
$$p'_y = p_y + t_y$$

- or in matrix form:

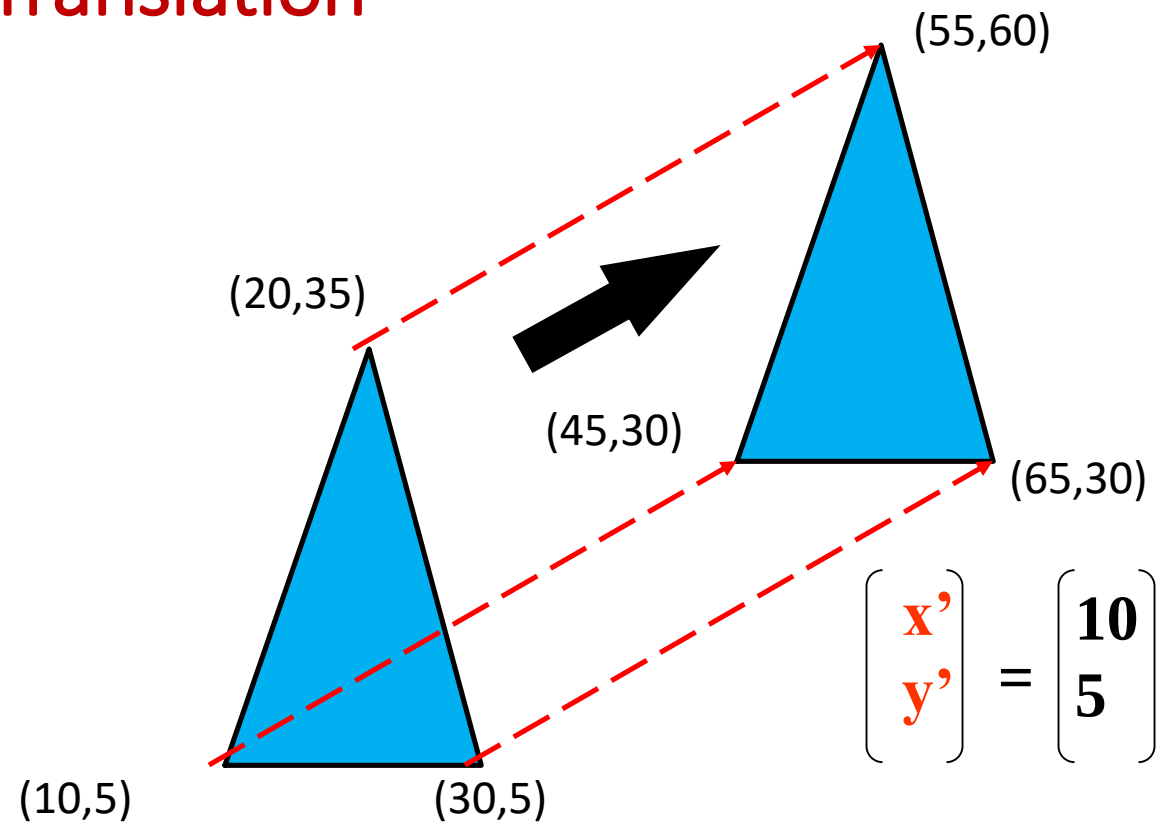
$$\mathbf{P'} = \mathbf{P} + \mathbf{T}$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$



Adding or subtracting a value to or from a coordinate translates it in space.

Translation



$$x' = x + t_x$$

$$y' = y + t_y$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \end{pmatrix} + \begin{pmatrix} 35 \\ 25 \end{pmatrix} = \begin{pmatrix} 45 \\ 30 \end{pmatrix}$$

Specifying a 2D-Translation in OpenGL:

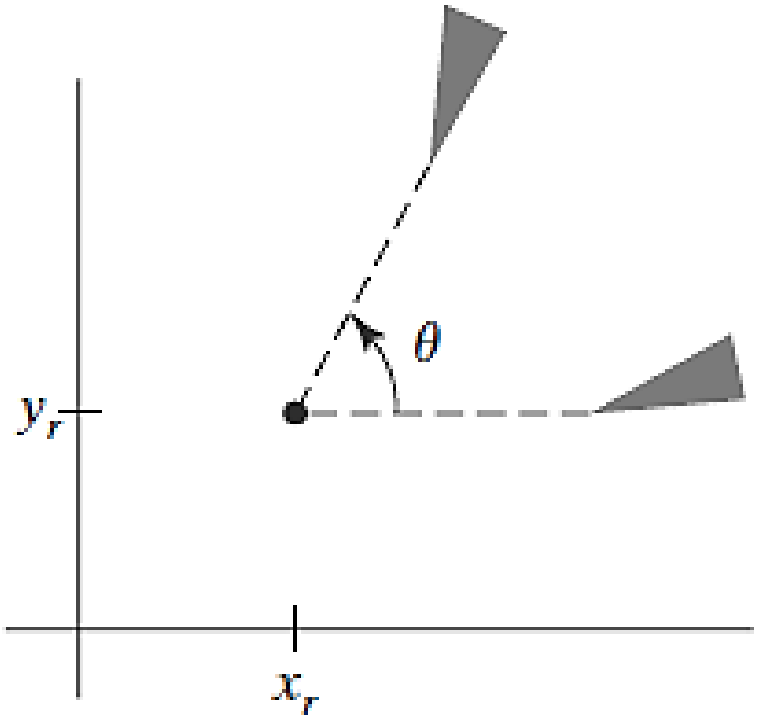
```
glTranslatef(tx, ty, 0.0);
```

(The z component is set to 0 in 2D translation

Rotation

- A rotation repositions all points in an object along a circular path in the plane centered at the pivot point.

- Parameters for the 2D rotation are the rotation angle θ and a position (x_r , y_r), called the rotation point (or pivot point.



Rotation

- Review Trigonometry

=> cos ϕ = **x/r** , sin ϕ= **y/r**

- **x = r.** cos ϕ**, y = r.**sin ϕ

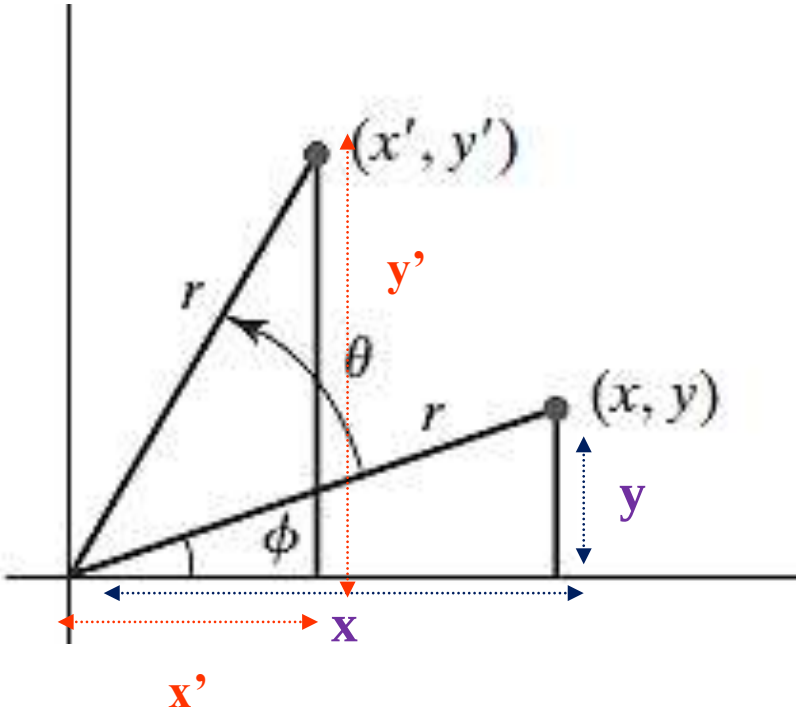
=> cos (ϕ+ θ) = x'/r

- x' = r. cos (ϕ+ θ)
- x' = r.cosϕcosθ - r.sinϕsinθ
- x' = x.cos θ – y.sin θ

=>sin (ϕ+ θ) = y'/r

y' = r. sin (ϕ+ θ)

- y' = r.cosϕsinθ + r.sinϕcosθ
- y' = x.sin θ + y.cos θ



Rotation

- We can write the components:

$$p'_x = p_x \cos\theta - p_y \sin\theta$$
$$p'_y = p_x \sin\theta + p_y \cos\theta$$

- or in matrix form:

$$P' = R \bullet P$$

- A positive value for the angle θ defines a counterclockwise rotation about the pivot point, as in our example, and a negative value rotates objects in the clockwise direction

- Rotation matrix

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Rotation

- Example
  - Find the transformed point, P’, caused by rotating P= (5, 1) about the origin through an angle of 90°.

Rotation

- Example **(Solution)**
  - Find the transformed point, P', caused by rotating P= (5, 1) about the origin through an angle of 90°.

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \bullet \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \cdot \cos\theta - y \cdot \sin\theta \\ x \cdot \sin\theta + y \cdot \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} 5 \cdot \cos 90 - 1 \cdot \sin 90 \\ 5 \cdot \sin 90 + 1 \cdot \cos 90 \end{bmatrix}$$

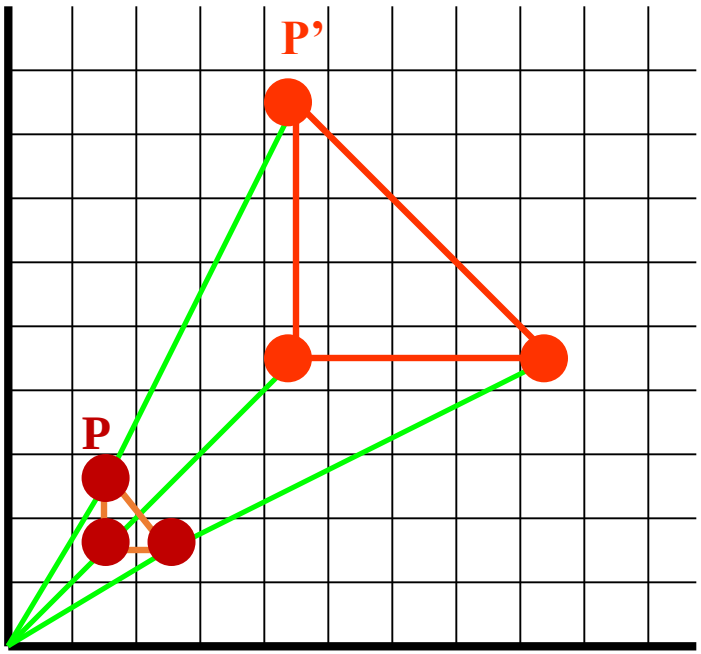
$$= \begin{bmatrix} 5 \cdot 0 - 1 \cdot 1 \\ 5 \cdot 1 + 1 \cdot 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 5 \end{bmatrix}$$

Scaling

- Scaling changes the size of an object and involves two scale factors, S_x and S_y for the x and y coordinates respectively.
- Scales are about the origin.
- We can write the components:

$$p'_x = s_x \bullet p_x$$
$$p'_y = s_y \bullet p_y$$



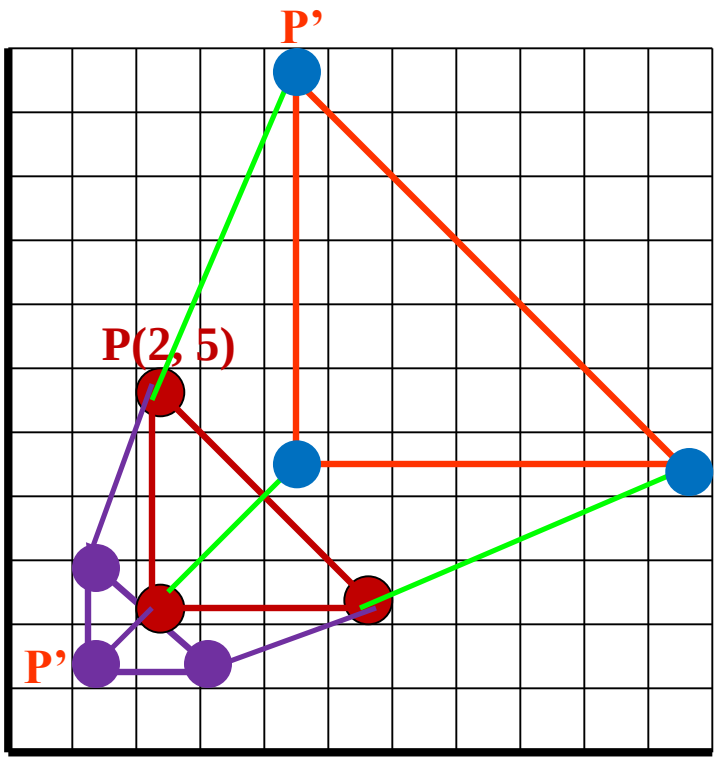
or in matrix form: **P' = S • P**

Scale matrix as:

$$S = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

Scaling

- If the scale factors are in between 0 and 1, the points will be moved closer to the origin ➔ the object will be smaller.
  - Example :
    - P(2, 5), Sx = 0.5, Sy = 0.5
    - Find P' ?
- If the scale factors are larger than 1, the points will be moved away from the origin ➔ the object will be larger.



- Example :
  - P(2, 5), Sx = 2, Sy = 2
  - Find P' ?

Scaling

- If the scale factors are the same, $S_x = S_y$ ➔ **uniform scaling** which maintains relative object proportions. Only change in size.
- If $S_x \neq S_y$ ➔ **differential scaling**. Change in size and shape

- Example : square → rectangle
  - P(1, 3), $S_x = 2$, $S_y = 5$, P' ?

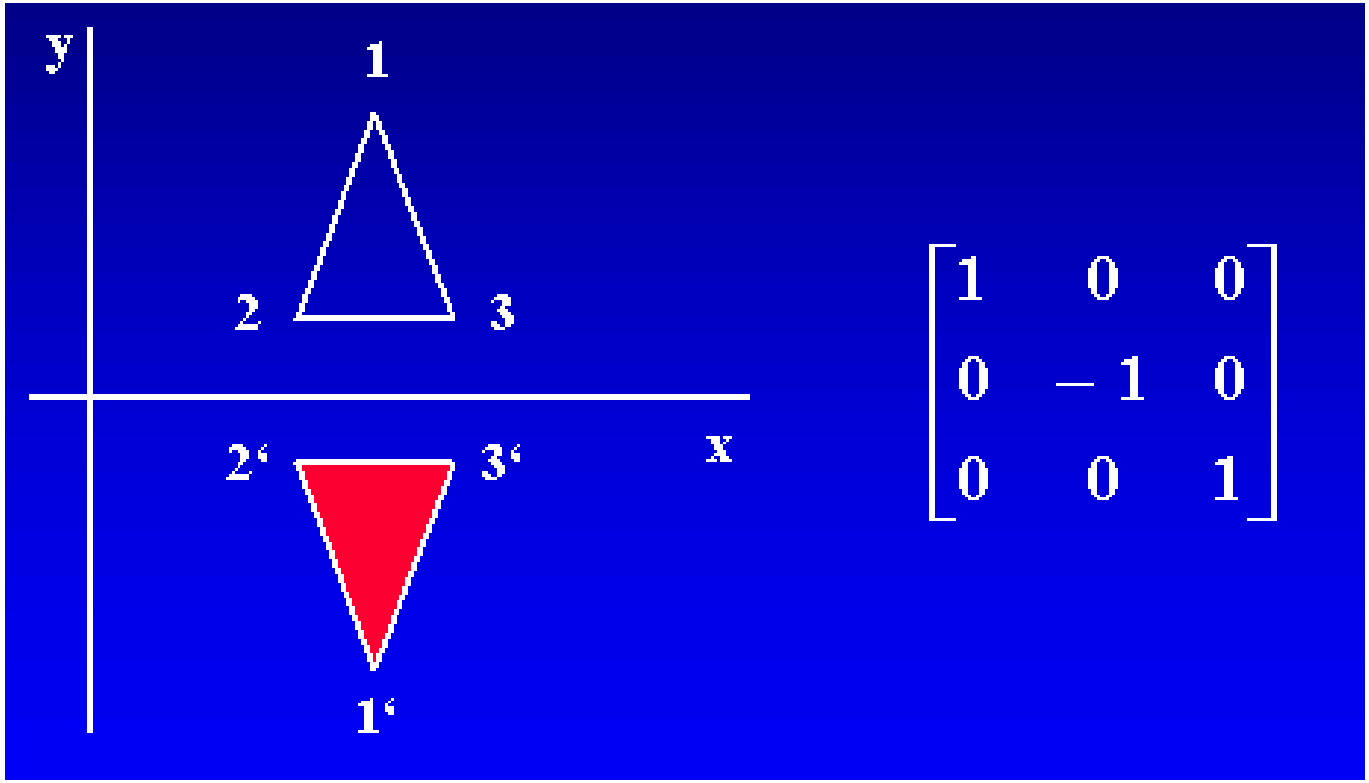
- In some systems, negative values can also be specified for the scaling parameters. This not only resizes an object, it reflects it about one or more of the coordinate axes

Reflection

- A transformation that produces a mirror image of an object.
- Axis of reflection
  - A line in the xy plane
  - A line perpendicular to the xy plane
  - The mirror image is obtained by rotating the object 180^0 about the reflection axis.
- Rotation path
  - Axis in xy plane: the rotation path about this axis is in a plane perpendicular to the xy plane.
  - Axis perpendicular to xy plane:the rotation path about this axis is in the xy plane.

Reflection

- **Reflection about the x axis**
- Reflection about the line y = 0
- It retains x values, but “flips” the y values of coordinate positions

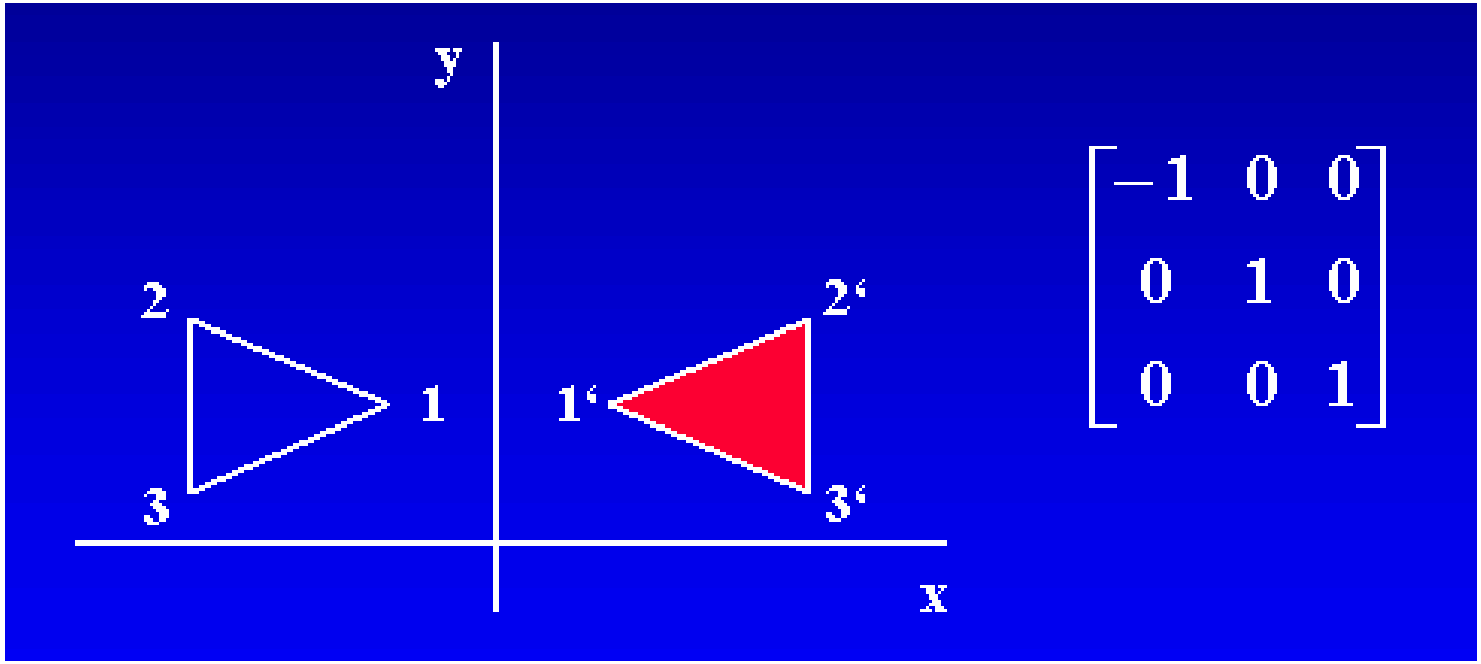


$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$y' = -y$$

Reflection

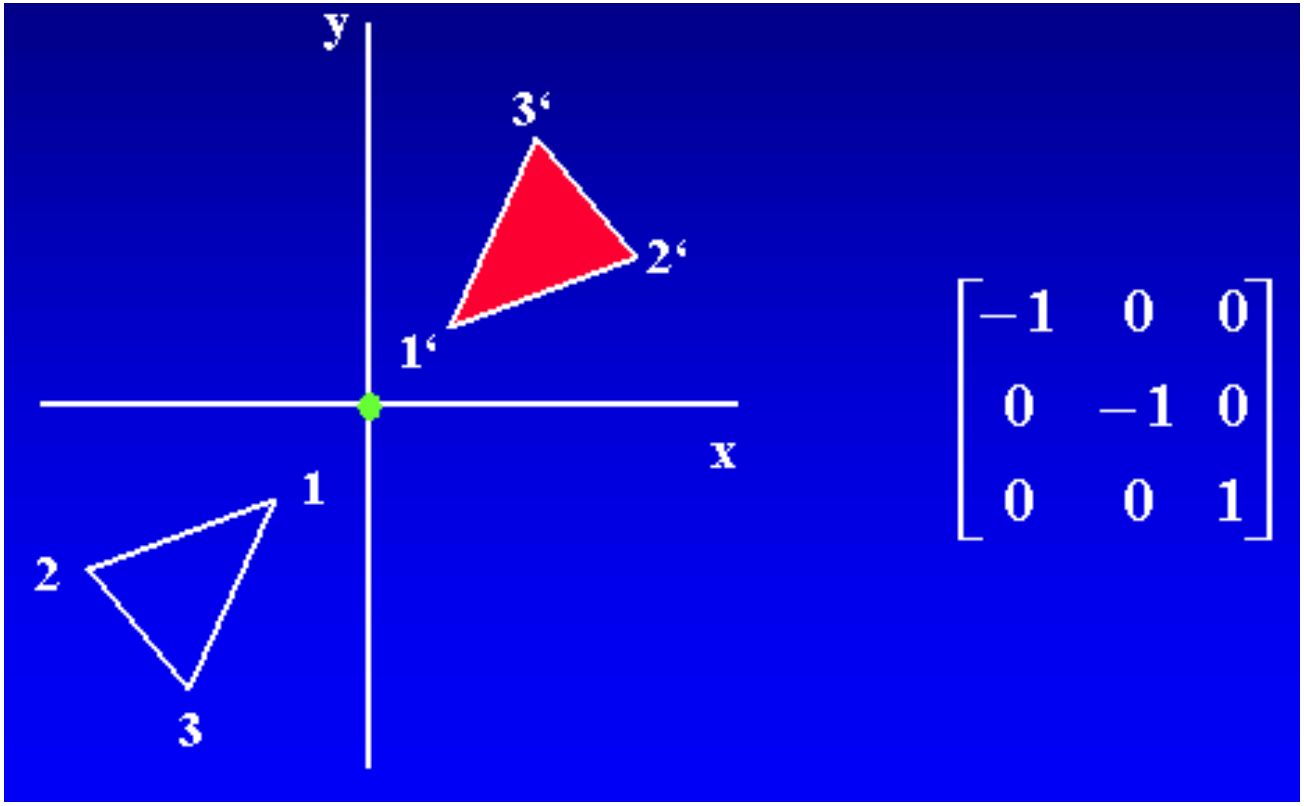
- **Reflection about the y axis**
- A reflection about the line x = 0 (the y axis)
- It flips x coordinates while keeping y coordinates the same



$x' = -x$

Reflection

- **Reflection relative to the coordinate origin**
- We flip both the x and y coordinates of a point by reflecting relative to an axis that is perpendicular to the xy plane and that passes through the coordinate origin

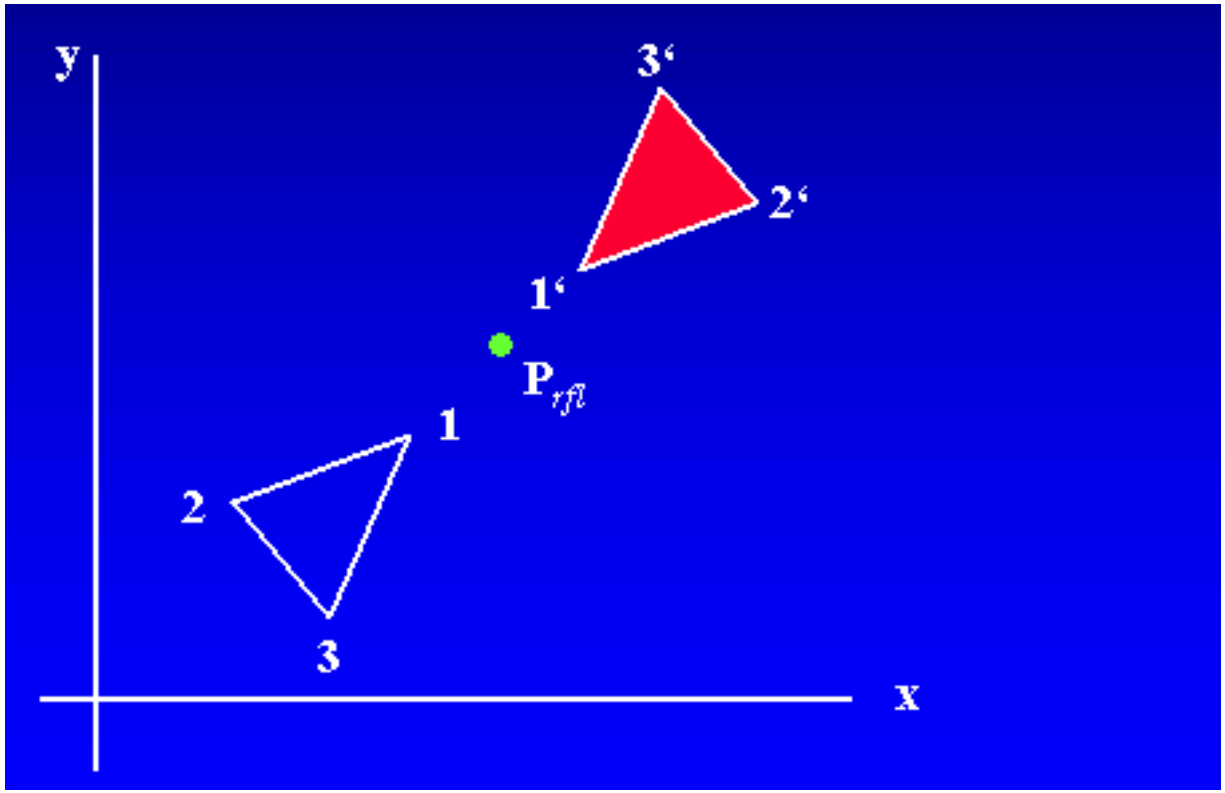


$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x' = -x$$
$$y' = -y$$

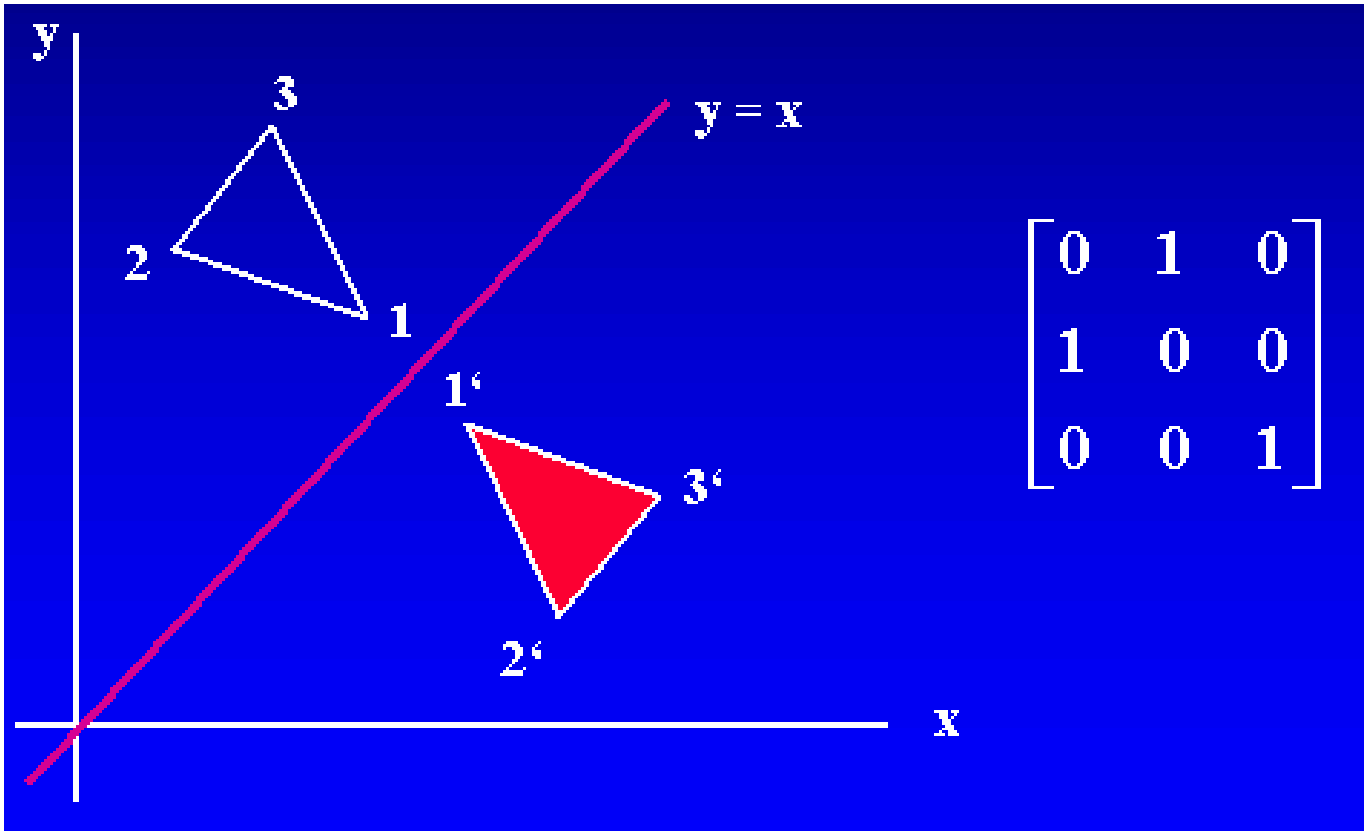
Reflection

- **Reflection of an object relative to an axis perpendicular to the xy plane through $P_{reflect}$**



Reflection

- **Reflection about the line y = x**

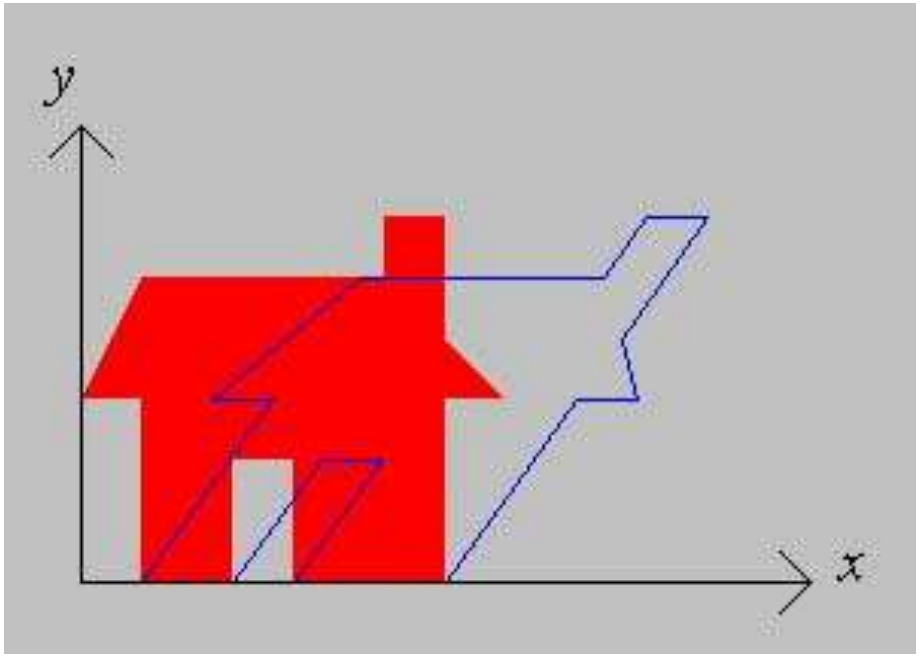


$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflections

Reflection about x:	`glScalef(1, -1, 1);`
Reflection about y:	`glScalef(-1, 1, 1);`
Reflection about origin:	`glScalef(-1, -1, 1);`

Shear



$$x' = x + h_x y$$

$$y' = y$$

- A shear transformation in the x-direction (along x) shifts the points in the x-direction proportional to the y-coordinate.

- The y-coordinate of each point is unaffected.

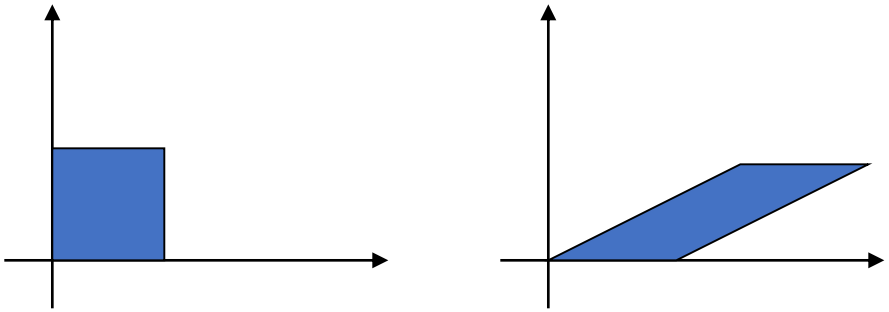
Shear

- **The x-direction shear relative to x axis**

$$\begin{bmatrix} 1 & sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x' = x + sh_x \cdot y$$
$$y' = y$$

If $sh_x = 2$:



General Properties

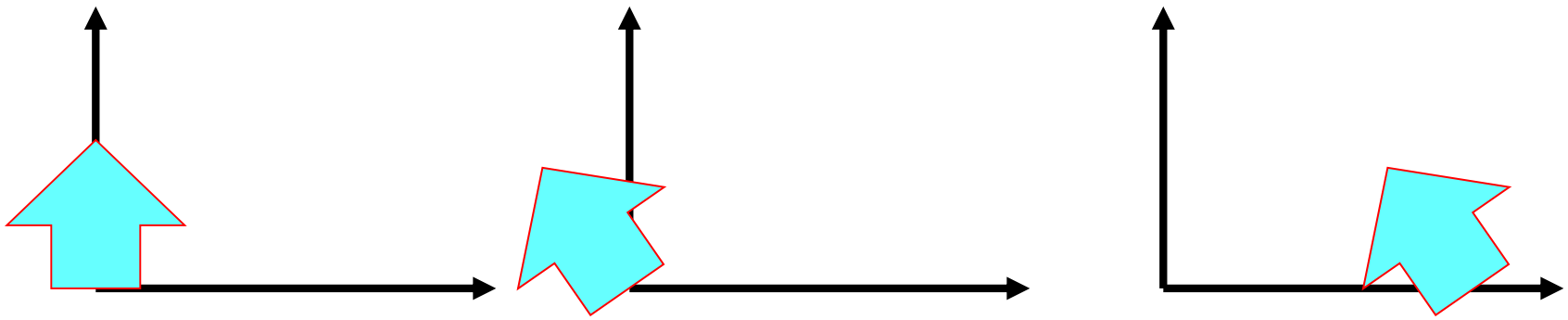
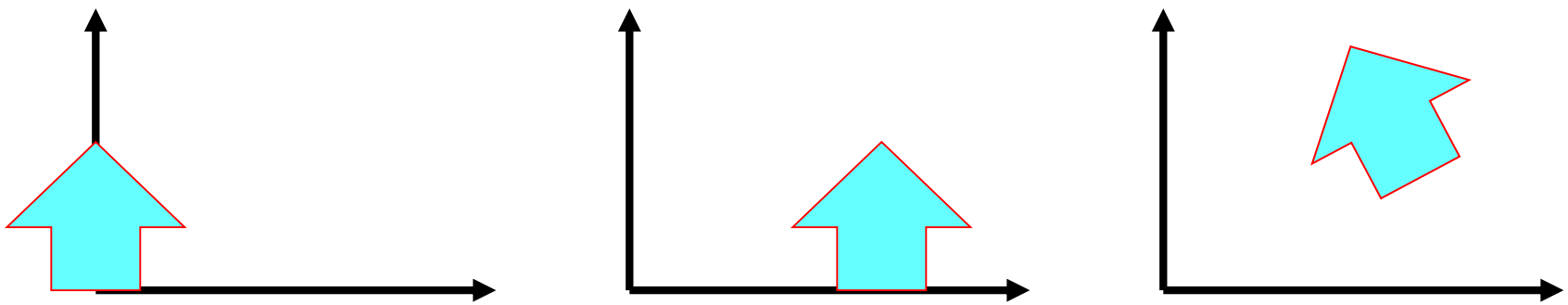
Preserved Attributes

	Line	Angle	Distance	Area
Translation	Yes	Yes	Yes	Yes
Rotation	Yes	Yes	Yes	Yes
Scaling	Yes	No	No	No
Reflection	Yes	Yes	Yes	Yes
Shear	Yes	No	No	Yes

Order of Transformations

In composite transformations, the order of transformations is very important.

Rotation followed by Translation:



Translation followed by Rotation: